

# Notes on Regular Epimorphisms and the Monadicity of the Pullback Functor in an Exact Category

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These notes are based on the work of Sobral and Tholen [ST92], with some ideas drawn from Janelidze and Tholen [JT91].

Let  $\mathcal{A}$  be a category with pullbacks and let  $p : E \rightarrow B$  be a morphism in  $\mathcal{A}$ . Consider the monad  $T = p^*p_!$  associated to  $p$ . Let  $(C, \gamma)$  be an object in  $(C \downarrow E)$  and let  $(C, \gamma; \xi)$  be an object in  $(\mathcal{A} \downarrow E)^T$  with  $\xi : T(C) \rightarrow C$  an algebra.

Consider the pullback in  $\mathcal{A}$ .

$$\begin{array}{ccc} E \times_B C & \xrightarrow{\text{proj}_2} & C \\ \text{proj}_1 \downarrow & \searrow \gamma & \downarrow p\gamma \\ E & \xrightarrow{p} & B \end{array}$$

Thus, we have the span in  $\mathcal{A}$ :

$$\begin{array}{ccc} & E \times_B C & \\ \text{proj}_2 \swarrow & & \searrow \xi \\ C & & C \end{array}$$

Since  $\xi$  is also a morphism in  $(\mathcal{A} \downarrow E)$ , we have that  $\gamma\xi = \text{proj}_1$

$$\begin{array}{ccc} E \times_B C & \xrightarrow{\xi} & C \\ & \searrow \text{proj}_1 & \swarrow \gamma \\ & E & \end{array}$$

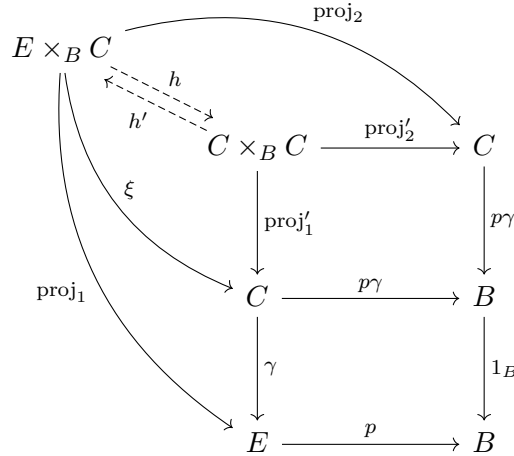
**Proposition 1.** *Let  $\mathcal{A}$  be a finitely complete category and let  $p : E \rightarrow B$  be a morphism in  $\mathcal{A}$ . For any object  $(C, \gamma; \xi)$  in  $(\mathcal{A} \downarrow E)^T$ , with  $T$  the monad associated to  $p$ . The pair  $(\text{proj}_2, \xi)$  is an equivalence relation in  $\mathcal{A}$*

*Proof.* First we show that  $(\text{proj}_2, \xi)$  is a regular epi is an internal relation. So we need

to show that  $\langle \text{proj}_2, \xi \rangle$  is a mono. Consider the following diagram of solid arrows<sup>1</sup>. It commutes, in fact

- $1_B(p\gamma)\text{proj}_2 = p\text{proj}_1 = p\gamma\xi$
- $1_B(p\gamma)\text{proj}'_2 = 1_B(p\gamma)\text{proj}'_1 = p\gamma\text{proj}'_1$

Thus we can consider the morphism  $h'$  induced by the universal property of the pullback from  $\text{proj}'_2$  and  $\gamma\text{proj}'_1$  and  $h$  similarly induced from  $\text{proj}_2$  and  $\xi$ .



In order to prove that  $\langle \text{proj}_2, \xi \rangle$  is a mono, we need to show that  $h$  is a mono. Since  $(\text{proj}_1, \text{proj}_2)$  is a jointly monic pair, by the following compositions

- $\text{proj}_1 h' h = \gamma \text{proj}'_1 h = \gamma \xi = \text{proj}_1 = \text{proj}_1 1_{E \times_B C}$
- $\text{proj}_2 h' h = \text{proj}'_2 h = \text{proj}_2 = \text{proj}_2 1_{E \times_B C}$

we have that  $h' h = 1_{E \times_B C}$ . This implies that  $h$  is a mono, in fact suppose that  $f$  and  $g$  are two morphisms such that  $hf = hg$ , then we have

$$\begin{aligned}
 hf &= hg \\
 \implies h' hf &= h' hg \\
 \implies 1_{E \times_B C} f &= 1_{E \times_B C} g \\
 \implies f &= g
 \end{aligned}$$

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<sup>1</sup>[ST92] assumes the existence of coequalizers, which is not the case in our setting. However, this is not a problem: at this point, we can use the morphism  $p\gamma$  instead of the coequalizer of  $(\text{proj}_2, \xi)$ , as it serves our purpose. Then, *a posteriori*, this proposition ensures, under the assumption of exact category, that the pair  $(\text{proj}_2, \xi)$  has a coequalizer, which we will make use of.

Consider now the canonical morphism  $i : C \times_B C \rightarrow C \times C$  induced by the universal property of the product. We have that  $i$  is a mono, in fact  $f$  and  $g$  are two morphisms such that  $if = ig$ , then we have

$$\begin{array}{ccccc}
 & & & C & \\
 & & \text{proj}'_1 \nearrow & \downarrow \pi_1 & \\
 A & \xrightarrow[f]{g} & C \times_B C & \xrightarrow{i} & C \times C \\
 & & \text{proj}'_1 \searrow & \uparrow \pi_2 & \\
 & & & C & 
 \end{array}$$

- $if = ig \implies \pi_1 if = \pi_1 ig \implies \text{proj}'_1 f = \text{proj}'_1 g$
- $if = ig \implies \pi_2 if = \pi_2 ig \implies \text{proj}'_2 f = \text{proj}'_2 g$

Since  $(\text{proj}'_1, \text{proj}'_2)$  is a jointly monic pair, we have that  $f = g$ . Thus,  $i$  is a mono. Since  $i$  and  $h$  are both monos, we have that  $ih$  is a mono.

By the following commutative diagram

$$\begin{array}{ccccc}
 & & \text{proj}_1 \nearrow & & \\
 & & & C & \\
 & & \text{proj}'_1 \nearrow & \downarrow \pi_1 & \\
 E \times_B C & \xrightarrow{h} & C \times_B C & \xrightarrow{i} & C \times C \\
 & & \text{proj}'_2 \searrow & \uparrow \pi_2 & \\
 & & & C & \\
 & & \text{proj}_2 \searrow & & 
 \end{array}$$

we conclude that  $ih = \langle \text{proj}_2, \xi \rangle$ , so  $\langle \text{proj}_2, \xi \rangle$  is a mono.

We proved that  $(\text{proj}_2, \xi)$  is an internal relation, now we need to show that it is an equivalence, ie it is reflexive, symmetric and transitive.

- Claim:  $(\text{proj}_2, \xi)$  is reflexive.

Called  $\eta$  the unit of the monad, we have that  $(\text{proj}_2, \xi)\eta_{(C, \gamma)} = \langle 1_C, 1_C \rangle$ . In fact  $\xi\eta_{(C, \gamma)} = 1_C$  and by the definition of the unit of the monad, we have that  $\text{proj}_2\eta_{(C, \gamma)} = \text{proj}_2\langle \gamma, 1_C \rangle = 1_C$ .

$$\begin{array}{ccccc}
 & C & & & \\
 & \swarrow 1_C & \downarrow \eta_{(C, \gamma)} & \searrow 1_C & \\
 C & \xleftarrow{\text{proj}_2} & E \times_B C & \xrightarrow{\xi} & C
 \end{array}$$

- Claim:  $(\text{proj}_2, \xi)$  is symmetric.

Consider the following diagram, where  $\text{proj}_1''$  and  $\text{proj}_2''$  are the projections of  $E \times_B C$ . Let  $l$  be the morphism induced by the universal property of the pullback from  $1_{E \times_B C}$  and  $\gamma \text{proj}_2$ , in particular it is the unique such that

$$\gamma \text{proj}_2 = l \text{proj}_1'' = l \text{proj}_1(1_E \times \text{proj}_2)$$

and

$$\begin{array}{ccccc}
 & & 1_{E \times_B C} = l \text{proj}_2'' & & \\
 & & \text{---} 1_{E \times_B C} \text{---} & & \\
 E \times_B C & \xrightarrow{\quad l \quad} & E \times_B (E \times_B C) & \xrightarrow{\quad \text{proj}_2'' \quad} & E \times_B C \\
 \downarrow \text{proj}_2 & & \downarrow 1_E \times_B \text{proj}_2 & & \downarrow \text{proj}_2 \\
 & & E \times_B C & \xrightarrow{\quad \text{proj}_2 \quad} & C \\
 & & \downarrow \text{proj}_1 & & \downarrow p\gamma \\
 C & \xrightarrow{\quad \gamma \quad} & E & \xrightarrow{\quad p \quad} & B
 \end{array}$$

$\downarrow 1_E \times_B \xi$        $\downarrow \xi$

Define  $\sigma = (1_E \times_B \xi)l$ . Observe that

$$\text{proj}_2 \sigma = \text{proj}_2 (1_E \times_B \xi)l = \xi 1_{E \times_B C} = \xi$$

It remains to show that  $\xi \sigma = \text{proj}_2$ . This holds because of the following compositions and the fact that  $(\text{proj}_1, \text{proj}_2)$  is a jointly monic pair.

- $\text{proj}_1(1_E \times_B \text{proj}_2)l = \text{proj}_1''l = \gamma \text{proj}_2$
- $\text{proj}_1 \eta_{(C, \gamma)} \text{proj}_2 = \text{proj}_1 \langle \gamma, 1_C \rangle \text{proj}_2 = \gamma \text{proj}_2$
- $\text{proj}_2(1_E \times_B \text{proj}_2)l = \text{proj}_2 \text{proj}_2''l = \text{proj}_2 1_{E \times_B C} = \text{proj}_2$
- $\text{proj}_2 \eta_{(C, \gamma)} \text{proj}_2 = \text{proj}_2 \langle \gamma, 1_C \rangle \text{proj}_2 = 1_C \text{proj}_2 = \text{proj}_2$

$$\begin{array}{ccccc}
 & & E \times_B C & & \\
 & \swarrow \xi & \downarrow \sigma = (1 \times_B \xi)l & \searrow \text{proj}_2 & \\
 C & & E \times_B C & & C \\
 & \nwarrow \text{proj}_2 & & \nearrow \xi & \\
 & & E \times_B C & & 
 \end{array}$$

- Claim:  $(\text{proj}_2, \xi)$  is transitive.  
Consider the pullback (???)

$$\begin{array}{ccc}
(E \times_B C) \times_C (E \times_B C) & \longrightarrow & E \times_B C \\
\downarrow & & \downarrow \xi \\
E \times_B C & \xrightarrow{\text{proj}_2} & C
\end{array}$$

□

In an exact category each equivalence relation is a kernel pair and by  $(-)$  has a coequalizer. Thus in an exact category  $(\text{proj}_2, \xi)$  has a coequalizer. Let  $q : C \rightarrow Q$  be such coequalizer. Observe that considering  $\text{proj}_2$  and  $\xi$  as morphism in  $(\mathcal{A} \downarrow B)$ , then  $q$  is a morphism in  $(\mathcal{A} \downarrow B)$  as well by the universal property. So, called  $\delta$  the morphism induced by the universal property of the coequalizer from  $\text{proj}_2$  and  $\xi$ , we have that  $q$  with codomain  $(Q, \delta)$  is also the coequalizer of the pair in  $(\mathcal{A} \downarrow B)$ .

$$\begin{array}{ccccc}
E \times_B C & \xrightarrow{\xi} & C & \xrightarrow{q} & Q \\
& \searrow \text{proj}_2 & \downarrow p\gamma & \swarrow \delta & \\
& & B & & 
\end{array}$$

$p\text{proj}_1$  (arrow from  $E \times_B C$  to  $B$ )

Consider the comparison functor  $K : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)^T$  defined as  $K(D, \zeta) = (p^*(D, \zeta), p^*(\epsilon_{(D, \zeta)}))$ , where  $\epsilon$  is the counit of the adjunction, defining  $\epsilon_{(D, \zeta)}$  to be the projection from  $E \times_B D$  to  $D$  of the pullback of  $p$  along  $\zeta$ .

In our context of exact category, turns out that the comparison functor  $K$  has a left adjoint  $L : (\mathcal{A} \downarrow E)^T \rightarrow (\mathcal{A} \downarrow B)$  defined as  $L(C, \gamma; \xi) = (Q, \delta)$ , where  $(Q, \delta)$  is the codomain of coequalizer of the pair  $(\text{proj}_2, \xi)$  in  $(\mathcal{A} \downarrow B)$ .

Let us introduce the counit  $\rho$  and counit  $\alpha$  of the adjunction  $K \dashv L$ . For each object  $(D, \zeta)$  in  $(\mathcal{A} \downarrow B)$ , the counit is a morphism of the kind  $\rho_{(D, \zeta)} : LK(D, \zeta) \rightarrow (D, \zeta)$ . First we compute  $K(D, \zeta)$ .

$$K(D, \zeta) = (E \times_B D, \pi_1; p^*(\pi_2))$$

$$\begin{array}{ccc}
E \times_B D & \xrightarrow{\pi_2} & D \\
\downarrow \pi_1 & & \downarrow \zeta \\
E & \xrightarrow{p} & B
\end{array}$$

Then, call  $q : E \times_B D \rightarrow Q$  the coequalizer of the equivalence relation  $(\text{proj}_2, p^*(\pi_2))$ , with codomain  $(Q, \delta)$  and compute  $L(K(D, \zeta))$ . We define in the diagram below the morphism

$\rho_{(D,\zeta)}$  by the universal property of the coequalizer.

$$\begin{array}{ccccc}
 E \times_B (E \times_B D) & \xrightarrow[pj_2]{p^*(\pi_2)} & E \times_B D & \xrightarrow{q} & Q \\
 \downarrow pj_2 & & \downarrow \pi_2 & \nearrow \rho_{(D,\zeta)} & \\
 E \times_B D & \xrightarrow{\pi_2} & D & & \\
 \downarrow \pi_1 & & \downarrow \zeta & \nearrow \delta & \\
 E & \xrightarrow{p} & B & & 
 \end{array} \tag{1}$$

Observe that  $\delta$  was also defined as the morphism induced by the universal property of the coequalizer, considering the object  $E \times_B D$  as an object in  $(\mathcal{A} \downarrow B)$  via  $p\pi_1$ . Thus, by the commutation of the diagram,  $\rho_{(D,\zeta)}$  is a morphism in  $(\mathcal{A} \downarrow B)$ .

(UNIT TO DO. Why are the two definitions of the unit  $\alpha$  the same? Does  $\xi$  coequalize  $(p^*(\xi), p^*(pj_2))$ )

Our goal is to show that  $p^*$  is a monadic functor, ie that the comparison functor  $K$  is an equivalence of categories. In order to do so, we aim to prove that the unit  $\alpha$  and the counit  $\rho$  of the adjunction  $K \dashv L$  are pointwise isomorphisms.

**Proposition 2.** *In an exact category if  $p : E \rightarrow B$  is a regular epi, then the counit  $\rho$  of the adjunction  $K \dashv L$  is a pointwise isomorphism.*

*Proof.* We aim to prove that for each object  $(D, \zeta)$  in  $(\mathcal{A} \downarrow B)$ , the morphism  $\rho_{(D,\zeta)} : LK(D, \zeta) \rightarrow (D, \zeta)$  is an isomorphism. Consider the diagram (1) of the definition of the counit  $\rho_{(D,\zeta)}$ . An exact category is also regular, in particular regular epi are pullback stable. If  $p$  is a regular epi, then  $\pi_2$  is a regular epi as well. The top left square is a pullback by (CITE), thus  $(p^*(\pi_2), pj_2)$  is the kernel pair of  $\pi_2$ . Since  $\pi_2$  is a regular epi, then, by (CITE), it is the coequalizer of its kernel pair  $(p^*(\pi_2), pj_2)$ . We have that  $q$  and  $\pi_2$  are coequalizers of the same pair, thus the morphism  $\rho_{(D,\zeta)}$  defined by the universal property of the coequalizer  $q$  is actually an isomorphism.  $\square$

## References

- [JT91] G. Janelidze and W. Tholen. “How algebraic is the change-of-base functor?” In: *Category Theory, proceedings of the International Conference held in Como, Italy, July 22-28, 1990*. Lecture Notes in Mathematics 1488. Berlin and Heidelberg: Springer, 1991, pp. 174–186.

- [ST92] M. Sobral and W. Tholen. “Effective descent morphisms and effective equivalence relations”. In: *Category Theory 1991, proceedings of an International Summer Category Theory Meeting held June 23-30, 1991*. Canadian Mathematical Society Conference Proceedings 13. Providence, RI: American Mathematical Society, 1992, pp. 421–433.